

Seungyong Moon

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Summary

Deep learning researcher with 7 publications at top-tier ML conferences, including 6 first-author papers and 800+ citations. Research spans reinforcement learning, self-supervised learning, and LLM reasoning, with strong interest in applying RL to sequential decision-making in real-world systems.

Education

- PhD** **Seoul National University**, Computer Science & Engineering Seoul, South Korea
 • Advisor: Prof. Hyun Oh Song Feb 2027
- BA/BS** **Seoul National University**, Economics/Mathematical Sciences Seoul, South Korea
 • Minor in Computer Science & Engineering Feb 2019
 • Summa Cum Laude (GPA: 4.10/4.30, ranked 1st out of 34 students)

Experience

- Qualcomm AI Research**, Research Intern Amsterdam, Netherlands
 • Developed an RL framework that trains LLMs to adaptively execute classical algorithms via a Python REPL based on input complexity, reducing execution steps and computational overhead while maintaining correctness. Sept 2024 – Jan 2025
 • Built a multi-turn RL fine-tuning pipeline for LLMs from scratch, enabling complex multi-step reasoning and tool-use interactions such support existed in mainstream libraries. 5 months
- KRAFTON AI**, Research Intern Seoul, South Korea
 • Developed a self-supervised RL algorithm for discovering hierarchical skills in a Minecraft-like game, achieving SOTA with 4% of the parameters of the previous best. June 2023 – Sept 2023
 • Designed an agentic framework that improves the spatial reasoning of GPT-4 in grid-based game environments, leveraging self-reflection on interaction history to infer environment dynamics. 4 months
- DeepMetrics**, Research Intern Seoul, South Korea
 • Built a data preprocessing pipeline for 20K+ hours of ventilator waveform data, enabling large-scale training of ventilator control policies. June 2022 – Sept 2022
 • Developed imitation learning, reinforcement learning, and off-policy evaluation methods for autonomous ventilator control. 4 months
- NAVER Search & Clova**, Research Intern Gyeonggi-do, South Korea
 • Developed a data augmentation method for paraphrase identification in seq2seq models, improving accuracy by 2%. July 2018 – Aug 2018
2 months

Publications

- Learning to Better Search with Language Models via Guided Reinforced Self-Training** NeurIPS 2025
 Seungyong Moon, Bumsoo Park, Hyun Oh Song
arxiv.org/abs/2410.02992
- Improving the Efficiency of Algorithmic Reasoning in Language Models via Reinforcement Learning** Preprint
 Seungyong Moon, Michaël Defferrard, Corrado Rainone, Roland Memisevic

Discovering Hierarchical Achievements in Reinforcement Learning via Contrastive Learning <i>Seungyong Moon</i> , Junyoung Yeom, Bumsoo Park, Hyun Oh Song arxiv.org/abs/2307.03486	NeurIPS 2023
Rethinking Value Function Learning for Generalization in Reinforcement Learning <i>Seungyong Moon</i> , JunYeong Lee, Hyun Oh Song arxiv.org/abs/2210.09960	NeurIPS 2022
Query-Efficient and Scalable Black-Box Adversarial Attacks on Discrete Sequential Data via Bayesian Optimization Deokjae Lee, <i>Seungyong Moon</i> , Junhyeok Lee, Hyun Oh Song arxiv.org/abs/2206.08575	ICML 2022
Preemptive Image Robustification for Protecting Users against Man-in-the-Middle Adversarial Attacks <i>Seungyong Moon</i> [*] , Gaon An [*] , Hyun Oh Song arxiv.org/abs/2112.05634	AAAI 2022
Uncertainty-Based Offline Reinforcement Learning with Diversified Q-Ensemble Gaon An [*] , <i>Seungyong Moon</i> [*] , Jang-Hyun Kim, Hyun Oh Song arxiv.org/abs/2110.01548	NeurIPS 2021
Parsimonious Black-Box Adversarial Attacks via Efficient Combinatorial Optimization <i>Seungyong Moon</i> [*] , Gaon An [*] , Hyun Oh Song arxiv.org/abs/1905.06635	ICML 2019

Awards and Scholarships

- NeurIPS Top Reviewers (2022, 2025)
- NeurIPS Scholar Award (2023)
- NAVER PhD Fellowship Award (2022)
- Youlchon AI Star Scholarship (2022)
- Qualcomm Innovation Fellowship Korea Finalists (2020, 2022)
- KFAS Computer Science Graduate Student Scholarship (2019 – 2024)

Skills

Languages: Python, C++, MATLAB

ML Frameworks: PyTorch, JAX, TensorFlow

LLM Libraries: Transformers, TRL, veRL, vLLM, SGLang

Infrastructure: Docker, Slurm

Patents

1. Apparatus and method with out-of-distribution data detection (US Patent Pub. 2024/0242072)
2. Image recognition method and apparatus, image preprocessing apparatus, and method of training neural network (US Patent 11,921,818)
3. Method and apparatus for classifying image (KR Patent 10-2505303)

Academic Services

- Conference Reviewer: NeurIPS (2021 – 2026), ICML (2022 – 2026), ICLR (2024 – 2026), AAAI (2022 – 2025), AAAI-AIA (2026), AAAI-Demo (2026), AISTATS (2025 – 2026), RLC (2024)
- Journal Reviewer: Neurocomputing (2021), Machine Learning (2023), Transactions on Intelligent Vehicles (2023)